Determining Dominant Charge Carriers and Band Gap Energy in Doped and Undoped Germanium Semiconductors

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INTRODUCTION

In 1879, Edwin Hall successfully measured a transverse voltage across a current-carrying strip of gold foil placed between two electromagnets. The magnetic field created by the electromagnets exerted a Lorentz force on moving charge carriers which caused them to accumulate on one side of the foil. After months of observation, Hall found that the separation of charge produced a measurable voltage perpendicular to the direction of the magnetic field and applied current. This phenomenon, in which a magnetic field induces a transverse voltage in a conducting material, is known as the Hall effect [1].

Although the Hall effect can be measured within any conducting material, such as the gold foil used by Hall, the high density of free electrons in these metals results in a much smaller Hall voltage. In semiconductors, however, due to the presence of a band gap, the ability to control the density of charge carriers results in a significantly larger and more easily measurable Hall voltage compared to conducting materials.

In this experiment, a current is applied to three different germanium samples in the presence of a magnetic field. Each sample differs in its dominant charge carrier: n-doped and p-doped samples contain an excess of electrons and holes, respectively, while the intrinsic (undoped) sample contains approximately equal concentrations of both. The Hall voltage and voltage along the direction of current are recorded as functions of current, temperature, and magnetic field, with certain measurements performed only for specific sample types. Ultimately, the influence of these different semiconductors on the Hall voltage is analyzed to determine the type, concentration, and mobility of dominant charge carriers within each sample. In addition, certain relations will be used to determine the band gap energy of the intrinsic germanium.

EXPERIMENTAL SETUP

This experiment tested three types of semiconductors, differing in their electrical properties: Ge-P (p-type), Ge-N (n-type), and Ge-Undoped (intrinsic). Each sample was inserted into the Hall effect console, with electromagnets positioned on either side. For each sample, the Hall voltage (V_H) and sample voltage (V_P) were recorded as functions of current, temperature, and magnetic field. The experimental procedures specific to each method and sample type are detailed in a later section.

RESULTS/ANALYSIS

The Lorentz force law describes the total force exerted on a charged particle by electric and magnetic fields:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \tag{1}$$

- $\vec{F}$ is the total force on the particle
- q is the electric charge of the particle
- $\vec{E}$ is the electric field
- $\vec{v}$ is the velocity of the particle
- $\vec{B}$ is the magnetic field

In the presence of a magnetic field, charged particles in motion will experience the magnetic component of force:

$$\vec{F}_m = q\vec{v}_d \times \vec{B} \tag{2}$$

- $\vec{v}_d$ is the drift velocity

When applying a magnetic field perpendicular to the current-carrying direction within a sample, charges will accumulate at the edge, creating an electric field.

$$\vec{F}_e = q\vec{E_H} \tag{3}$$

- $\vec{E_H}$ is the Hall electric field

This electric field creates a voltage difference along the direction perpendicular to both fields, known as the Hall voltage. Charges continue to accumulate until the electric force exactly balances the magnetic force on each charge carrier ($\vec{F}_m = \vec{F}_e$), resulting in zero net force:

$$q\vec{E_H} = q\vec{v}_d \times \vec{B} \tag{4}$$

The voltage difference produced when a magnetic field is applied to a current-carrying semiconductor is the basis of the Hall effect. This effect is employed in this experiment to analyze germanium samples of n-doped, p-doped, and intrinsic material, beginning with identifying the dominant charge carriers for each sample.

Verifying Dominant Charge Carriers

Electrons are negative charge carriers defined to have a charge of -1.602×10^{-19} C. Holes, while not real particles, exist as the absence of an electron in the valence band and are therefore treated as positive charge carriers

with the same magnitude of charge as an electron. Because the Lorentz force depends on the sign of the charge, positive and negative charges are deflected in opposite directions, resulting in a reversal of the electric field and the sign of the Hall voltage V_H. For a given sample, V_H is given by:

$$V_H = \frac{IB}{nqd} \tag{5}$$

- I is the current through the sample
- n is the density of the dominant charge carrier
- d is the thickness of the sample

To verify the dominant charge carriers within each sample, the current through the doped samples was first varied from -25 mA to +25 mA, with the Hall voltage recorded at 5 mA steps. This was repeated for the intrinsic sample using 1 mA steps within a range of -3 mA to +3 mA. As shown in Fig. 1, all three samples exhibit a linear relationship between Hall voltage and current. The n-type and p-type samples have oppositely signed slopes, while the intrinsic sample shows a slope of zero.

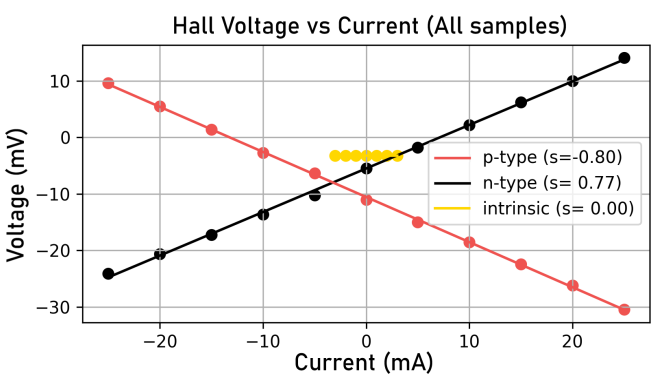

FIG. 1. Plot shows Hall voltage V_H as a function of sample current for the n-type, p-type, and intrinsic semiconductor. For the intrinsic sample, a value of V_H = -3.2 mV was obtained for each current reading, likely from a systematic offset.

Disclaimer: The germanium samples used in this experiment were designed with a reversed contact orientation, which effectively swapped sign conventions for the Hall voltage. This results in slopes oppositely signed relative to the dominant charge carrier of the material, causing a lot of unnecessary confusion.

The Hall voltage V_H measured for the p-type sample was oppositely signed relative to the sample current I, indicating that the dominant charge carriers in this sample are positively charged. Since conventional current is defined as the direction in which holes move, the signage of V_H directly correlates with the direction of current I, confirming holes as the dominant charge carriers in the p-type sample.

For the n-type sample, the sign of V_H was consistent with the sign of the sample current, indicative of a material dominated by negative charge carriers such as holes. This relationship is characteristic of a material dominated by electrons, which carry a negative charge and move opposite the direction of conventional current.

Varying Magnetic Field Strength

The strength of the magnetic field was varied to observe its effect on the Hall voltage produced for the n-type and p-type samples. Because previous variations in current for the intrinsic sample in the presence of a magnetic field did not produce a measurable Hall voltage, varying the strength of the magnetic field for this sample would also not yield a detectable voltage.

For the n-type and p-type samples, the current generating the magnetic field was varied from 0 A to 7 A, measuring the Hall voltage in 1 A steps. The relationship between V_H and B is shown in Fig. 2. Increases in the magnetic field strength proportionately scale the magnitude of the Hall voltage, with the slope being consistent with the type of dominant charge in each sample. Since the Lorentz force acting on charge carriers is directly proportional to the magnetic field strength, this proportional increase in the Hall electric field is expected.

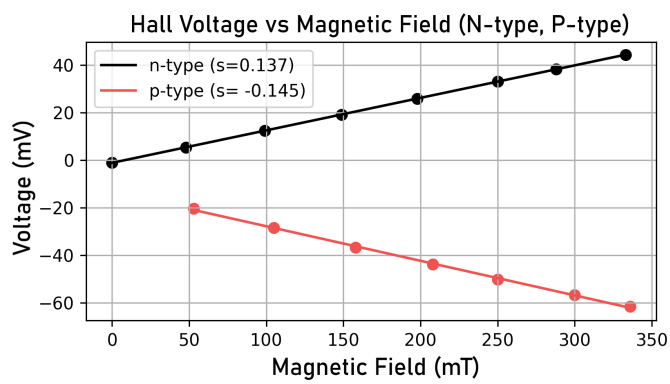

FIG. 2. Plot shows Hall voltage as a function of magnetic field strength for the n-type and p-type samples. The slopes inversely correlate with the sign of the dominant charge.

The sample voltage V_P was also measured for the doped samples while similarly varying the magnetic field. Since the sample voltage is taken along the direction of current, it should be approximately independent of the magnetic field. Fig. 3 shows the relationship between V_P and B, with V_P remaining relatively constant despite increases in magnetic field. This constant relationship confirms that variations in previous measurements of V_H vs I are independent of the magnetic field, and are unlikely to result from current or resistance drift caused by the magnetic field strength.

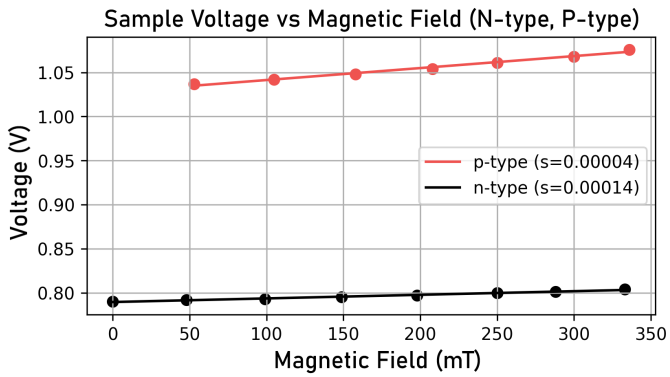

FIG. 3. Plot shows sample voltage as a function of magnetic field strength for the n-type and p-type samples. The near constant slopes verify that noise is unlikely to result from the effect of the magnetic field on the sample resistance.

Computing Charge Carrier Density

After verifying the dominant charge carriers for each sample, the density n was determined using the slopes V_H/I obtained from the regression shown in Fig. 1. The equation can also be rewritten using the Hall constant R_H, which relates V_H to the type and density of charge carriers in a material.

$$n = \frac{I}{V_H} \cdot \frac{B}{qd} = \frac{1}{qR_H} \tag{6}$$

- R_H is the Hall constant

The values of n represent the number of mobile charge carriers per unit volume. Table 1 shows the number of electrons and holes per cubic meter for the n-type and p-type samples, respectively, calculated using a magnetic field $B = 1$ A and sample thickness $d = 1$ mm. The Hall voltage is insufficient for determining the carrier density in the intrinsic sample, due to its nearly equivalent concentration of both holes and electrons, resulting in no detectable change in V_H for variations in current.

Sample	**Slope** V_H/I	**Density** n **(m^{-3})**	R_H **(m$^3/C$)**
p-type	-0.80	7.8×10^{21}	-8.0×10^{-4}
n-type	0.77	8.1×10^{21}	7.7×10^{-4}
intrinsic	0.00	-	-

TABLE I. Dominant charge carrier density for each sample. The density and Hall constant could not be determined for the intrinsic sample due to an equal concentration of both charges.

Determining Conductivity and Mobility

Conductivity is the measure of a material's ability to conduct electricity. It is proportionally related to the reciprocal of resistivity (and inversely related to resistance for fixed geometry), and is given by:

$$\sigma = \frac{L}{RA} \tag{7}$$

- L is the length along the current direction
- R is the resistance
- A is the cross-sectional area

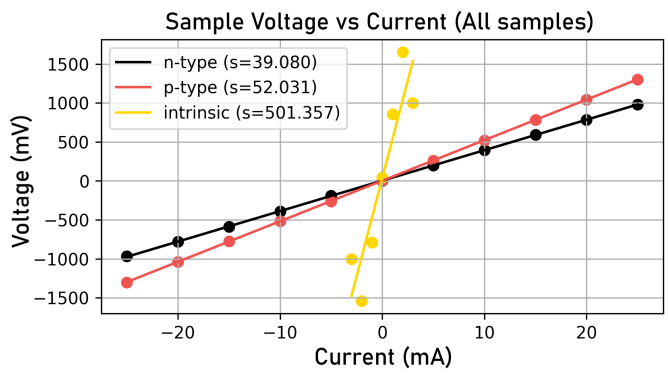

FIG. 4. Plot shows Sample voltage V_P as a function of sample current for the n-type, p-type, and intrinsic semiconductor. The slope V/I represents the sample resistance R.

To determine the conductivity for each sample, the cross-sectional area A is calculated using width $w = 10$ mm and thickness $d = 1$ mm. The resistance R was determined by plotting sample voltage V_P as a function of sample current I (Fig. 4), then applying Ohm's law to equate the slope V_P/I to R. The conductivity was computed using Eq. 7 for a sample of length $L = 20$ mm.

Sample	**Resistance** (V_P/I)	**Conductivity**	**Mobility**
n-type	39	51.3	-410.4
p-type	52	38.5	296.5
intrinsic	501	3.99	-

TABLE II. Computed conductivity σ (m$^3/C$) and mobility μ (cm$^2/V \cdot s$) for each sample. μ could not be computed for the intrinsic sample due to an undetermined Hall constant R_H.

Mobility μ is a property of the material that depends on factors such as temperature, density, and doping levels. It is directly related to conductivity by:

$$\mu = \frac{\sigma}{nq} = \sigma R_H \tag{8}$$

With the Hall constants R_H and charge carrier densities previously determined using Eq. 6, the mobility was computed for the n-type and p-type samples, as shown in Table II. Since the Hall constant for the intrinsic sample could not be determined with meaningful accuracy, the carrier density and, subsequently, mobility also remain undefined for this sample.

Varying Temperature

As shown previously, the density of charge carriers and dimensions of the material are notable sample properties that influence its conductivity and electrical behavior. Despite conductivity depending on both carrier density and mobility, mobility itself is an intrinsic material property determined by the scattering of charge carriers.

However, the mobility in a material can also be affected by external conditions such as its temperature, which can affect the efficiency at which charge carriers move through a material. Fig. 5 shows the relationship between Hall voltage and temperature, used to contrast the effects of temperature variations on concentration and mobility between doped and undoped samples.

The intrinsic sample exhibits a weak dependence on temperature, as the Hall voltage remains unchanged despite temperature variations due to the Hall effect being canceled by the balance of electrons and holes within the material. This results in the flat curve shown in Fig. 5.

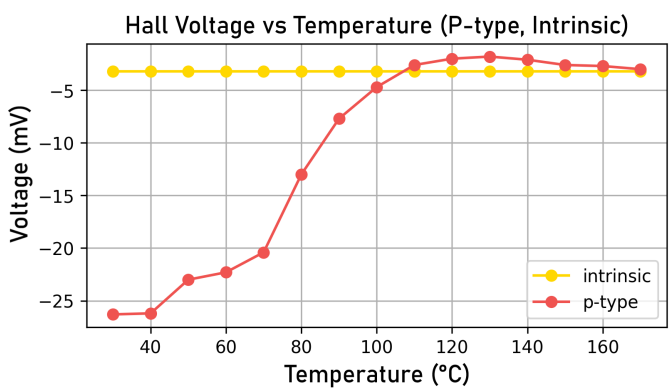

FIG. 5. Plot shows Hall voltage V_H as a function of sample temperature for the p-type and intrinsic semiconductor.

The p-type sample, dominated by holes, exhibits different behaviors at different temperature ranges. The Hall effect is the most detectable when temperature is low due to the excess amount of holes obtained from doping. As temperature increases into the mid-range, electrons within the material begin acquiring sufficient energy to jump the band gap and contribute to current, leaving a hole in their absence, and effectively forming an electron-hole pair. This leads to an increase in conductivity, but also a diminished Hall effect. At high temperatures, since the increased number of electrons reduces the dominance of holes, the p-type material exhibits a similar curve to the intrinsic sample. Despite its initial doping, it transitions to intrinsic conduction at high temperatures.

Determining the Band Gap Energy

To further contrast the effect of temperature T on conductivity σ between doped and undoped materials, Fig. 7 was constructed showing the relationship $\ln(\sigma)$ vs 1/T for the p-type and intrinsic sample. The intrinsic sample exhibits a linear relationship consistent with expectation. In contrast, the p-type sample deviates from this linearity at lower temperatures, and becomes linear and parallel to the intrinsic sample at higher temperatures.

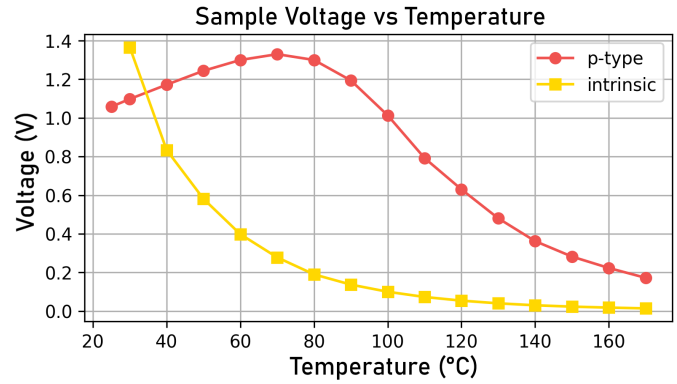

FIG. 6. Plot shows sample voltage V_H as a function of sample temperature for the p-type and intrinsic semiconductor.

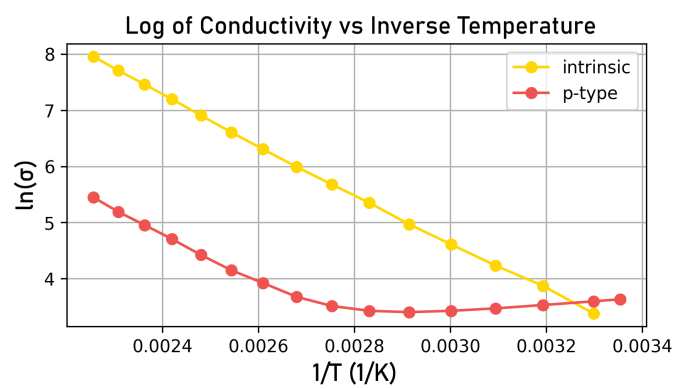

FIG. 7. Plot shows natural log of conductivity as a function of inverse temperature for the p-type and intrinsic sample.

A semiconductor consists of a region of forbidden energy which electrons are unable to occupy, known as the band gap. The dependence of conductivity on temperature is governed by the excitations of electrons across this gap, or from the valence band to the conduction band. Crossing the band gap requires that the electrons gain a sufficient amount of energy. This energy condition is fulfilled by raising the temperature of the material, resulting in more electrons crossing the gap and subsequently increasing the number of charge carriers. This behavior is described by [2]:

$$\sigma = A\exp(-\frac{E_g}{2kT}) \tag{9}$$

- E_g is the band gap energy
- k is the Boltzmann's constant

- T is the temperature

By taking the natural log of this equation, we find:

$$\ln(\sigma) = \ln(A) - \frac{1}{T}\frac{E_g}{2k} \tag{10}$$

which has the form of a linear equation in 1/T. Plotting $\ln(\sigma)$ vs 1/T produces a straight line with the slope $m = -E_g/2k$. Using the slope m = -4377 obtained from the intrinsic sample in Fig. 7, the band gap energy was calculated:

$$E_g = -2k \cdot m = -2(8.617 \times 10^{-5}) \cdot (-4377) = \mathbf{0.75\ eV}$$

The energy needed to move an electron from the valence band to the conduction band of the intrinsic germanium semiconductor material is 0.75 eV.

Above and Beyond

The plot of $\ln(\sigma)$ vs 1/T in Fig. 7 was analyzed in a physical context with respect to both the p-type and intrinsic sample in combination. The initial slopes of the two samples are proportional to the band gap energy E_g, and remain parallel and linear at high temperatures. This indicates that at high temperatures, conduction for both samples is dominated by intrinsic carriers, with both samples being governed by the same band gap energy, as predicted by the relation m = -$E_g/2k$.

As temperature decreases to the mid-range, the p-type sample begins to deviate from this linearly due to conduction being dominated by the positive charge carriers, rather than intrinsic conduction.

At a certain point in the low temperature range, both the doped and undoped sample exhibit similar conductivities that arise from completely different mechanisms. The intrinsic sample remains dominated by charge carriers generated from thermal excitation, while the p-type sample exhibits conduction resulting from the excess holes obtained from doping.

CONCLUSION

The n-type germanium sample was found to have a charge carrier density 7.8×10^{21} m^{-3} and Hall constant of -8.0×10^{-4} m^3/C. Using the resistance along the path of current, the material was found to have a conductivity of 51.3 S/m and mobility -410.4 cm^2/V·s. From this, the dominant charge carrier for this material was confirmed to be the negatively-charged electron.

The p-type germanium sample was found to have a charge carrier density 8.1×10^{21} m^{-3} and Hall constant of 7.7×10^{-4} m^3/C. The material was found to have a conductivity of 38.5 S/m and mobility 296.5 cm^2/V·s. From this, the dominant charge carrier for this material was confirmed to be the positively-charged hole.

Further confirmation of charge carrier dominance was obtained from comparison of the p-type and n-type mobility. Because electrons move freely in the conduction band while holes are restricted to the valence band, a larger magnitude of mobility for the n-type sample is expected compared to the p-type sample. This aligns with results from the experiment.

The charge carrier density, Hall constant, and mobility could not be determined for the intrinsic/undoped germanium sample due to the balance of electrons and holes resulting in an undetectable Hall voltage. The conductivity was determined to be 3.99 S/m, a small value relative to that of doped samples. This is consistent with the increased conductivity of doped samples resulting from the presence of dominant charge carriers.

Using the relationship for log of conductivity vs inverse temperature, the band gap energy of the intrinsic germanium sample was found to be 0.75 eV, which is in reasonable agreement with the accepted value of approximately 0.66 eV for germanium at 300 K [3].

The uncertainty of the dimensions of the semiconductor material was a potential source of error that may have led to over/underestimation of the conductivity. Measurements taken with respect to the sample temperature may also vary slightly as temperature adjustment was done using an analog switch that swept the full temperature range at different speeds.

Overall, the type of dominant charge carriers in the n-type and p-type germanium samples were successfully verified from determination of the charge carrier density and Hall constant. The conductivity and mobility for these samples were also found to align with the expected behavior. Due to a balance of positive and negative charge carriers, the charge carrier density and Hall constant could not be determined for the intrinsic germanium sample. Nevertheless, the relationship between the log of conductivity and inverse temperature allowed for determination of the band gap energy for this material.

AUTHOR CONTRIBUTIONS

S.T. wrote the introduction and experimental setup, G.D. and S.T. wrote the results and conclusion.

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